

RESEARCH ARTICLE

Fast backward steps and detachment of single kinesin molecules measured under a wide range of loads

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Funding information

MEXT, Grant/Award Numbers: 19H03189, JP16H04773

Abstract

To understand force generation under a wide range of loads, the stepping of single kinesin molecules was measured at loads from -20 to 42 pN by optical tweezers with high temporal resolution. The optical trap has been improved to halve positional noise and increase bandwidth by using 200 -nm beads. The step size of the forward and backward steps was 8.2 nm even over a wide range of loads. Histograms of the dwell times of backward steps and detachment fit well to two independent exponential equations with fast (~ 0.4 ms) and slow (> 3 ms) time constants, indicating the existence of a fast step in addition to the conventional slow step. The dwell times of the fast steps were almost independent of the load and ATP concentration, while those of the slow backward steps and detachment depended on those. We constructed the kinetic model to explain the fast and slow steps under a wide range of loads.

KEYWORDS

force measurement, kinesin, reaction model, single molecules

1 | INTRODUCTION

A team of kinesin and dynein molecules works in vesicle transport and at mitosis in cells. Kinesin possibly moves under a large hindering force during a tug-of-war between a few molecules of kinesin and a large number of dynein molecules.^{1,2} Some kinesin molecules in the kinesin assembly in vesicle transport and mitosis will push the other kinesin molecules forward, as myosin does in muscle.³ In mitosis, many species of kinesin superfamily proteins work at one time.⁴ The slow-moving kinesin among them will be stretched forward with an assisting load by fast-moving kinesin. Therefore, kinesin in cells works at a wide range of loads from assisting loads to large hindering loads.

The forward step of kinesin was detected in a narrow range of loads (2 – 7 pN) at the early stages of single molecule studies.^{5–7} In subsequent investigations, the backward steps and detachment in addition to the forward steps were analyzed near no load to stall load.^{8,9} The load range of the forward and backward steps was

expanded to -15 to 15 pN.^{10,11} The dwell time of the forward step was close to that of the backward step, indicating that both steps start from a common chemical state as a branched chemical reaction.^{8,9} The dwell times of forward steps were almost constant at an assisting load from -1 to -15 pN. However, these dwell times could have been overestimated because mean dwell time was calculated from data where undetectably short dwell times $< \sim 4$ ms were not included.¹⁰ Therefore, it is crucial to measure the step size and dwell times with high temporal resolution to understand the load dependency of stepping over wide load ranges.

The backward step was observed at the hindering load but was not detected at the assisting load.¹⁰ One possibility for no detection is that a backward step rarely occurred. The recent model calculation on the forward and backward steps predicted that the occurrence ratio of the backward step at the assisting load (-10 to 0 N) is $< 1\%$ of the total steps.¹² If we measure hundreds of steps at the assisting load, then a few backward steps will appear. The model also predicted that

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the dwell time of the backward step is shorter than that at no load. To detect the backward steps at the assisting load, we need to measure hundreds of steps at high temporal resolution.

The step size of kinesin is predominantly 8 nm at both forward and backward steps.^{5,6,8,10,13} The 16-nm steps in the backward direction were observed at 3–8 pN, and the occurrence rate of 16 nm among all backward steps was 10%–15%.^{8,14} The 16-nm steps in the backward direction were also observed at superstall loads.¹⁰ It is unknown whether the 16-nm step consists of double steps of fast and slow 8-nm steps or slippage.^{8,14} High temporal resolution is again required to determine whether kinesin takes 16-nm steps or double 8-nm steps.

Here, to understand the stepping reaction of kinesin over a wide range of loads, we detected the forward and backward steps and detachment at loads of –20 to 42 pN with high temporal resolution (~0.1 ms). We measured the dwell time of the steps and detachments and the stepping ratio of the number of backward steps to that of forward steps. We found a fast backward step and fast detachment with a dwell time of ~0.4 ms in addition to the normal slow steps. The dwell time of forward steps at the assisting load and backward steps under the superstall force (> 30 pN) was shorter than that under no load.

2 | RESULTS

2.1 | Forward step, backward step and detachment under a wide range of loads

We measured the stepping motion of single molecules of kinesin under a wide range of loads from –20 to 42 pN. Assisting (<0 pN) and superstall loads (>9 pN) were applied to kinesin beads by shifting the trap position toward the plus and minus end directions, respectively, of microtubules when the kinesin force reached ~2 pN (Figure 1A). We fixed the trap position at a force lower than the stall force (middle in Figure 1A). Kinesin beads showed forward steps, backward steps and detachments. Under an assisting load, forward steps were predominantly observed, and backward steps and detachment were rarely observed (inset in Figure 1A). Under a zero-to-stall load, kinesin took forward and backward steps and finally detached from microtubules after generating the maximum force of 7–8 pN, as previously reported.^{6,7} Under a superstall load >16 pN, only backward steps and detachments were observed (right trace in Figure 1A).

The step size was determined by the step-finding algorithm. The step sizes of the forward steps were predominantly 8 nm in all load ranges (Figure 1B). The step size of the backward step was –8 and –16 nm at low temporal resolution (bandwidth of 1 kHz) (Figure S1). However, most steps at –16 nm were resolved into sequential fast and slow steps with step sizes of –8 nm at high temporal resolution of 50–100 μ s (inset in Figure 1A). It was confirmed that the performance of step-finding algorithm is higher than ~80% at the temporal resolution of 50–100 μ s (see Section 4 and Figure S6). The mean step size in all ranges of loads was 8.17 ± 0.15 nm (mean $\pm$ SE) (Figure 1D).

To understand the load-dependent kinetics of single molecules of kinesin, the mean dwell times (τ) were calculated. The histograms

of the dwell times (t) of forward steps (–20 to –1 and 1 to 8 pN), backward steps (–7 to –1 pN) and detachment (–20 to –1 pN) were fitted well to the following Equation (1) with a single exponential term:

$$N_1(t) = N_i \left[1 - \exp\left(-\frac{t}{\tau_i}\right) \right] - n_i \quad (i=f, b, d) \quad (1)$$

where N_i is the number at infinite t , τ_i is the time constant and n_i is the number of undetectable steps because of the very short dwell time (Figure 2). The mean dwell time of the forward step decreased gradually with increasing assisting load (Figures 2A and 3A), indicating that kinesin's stepping motion was accelerated by the assisting load. The mean dwell time of the backward step and detachment under the assisting load was almost the same as that of the forward step, while the backward step was seldom observed (Figures 2B,C and 3A,B).

Under hindering loads from ~1 to 42 pN, the cumulative number of dwell times of the backward steps and detachments had a fast population at <1 ms and then a slow population (Figure 2B,C). Thus, the cumulative number (N_2) was not fitted to Equation (1) but was fitted to the following Equation (2) with slow and fast time constants (τ_i , τ_{if})

$$N_2(t) = N_i \left[1 - \exp\left(-\frac{t}{\tau_i}\right) \right] + N_{if} \left[1 - \exp\left(-\frac{t}{\tau_{if}}\right) \right] - n_i \quad (i=b \text{ and } d), \quad (2)$$

where N_i and N_{if} are the numbers at infinite t , n_i is the number that corresponds to undetectable steps and τ_{if} < 2 ms, and suffix i indicates backward step (b) and detachment (d). This indicates that the fast reaction is a process independent of the slow one. The slow mean dwell times (time constant, τ_{bf} in Equation 2) of the backward step increased with increasing load up to 8 pN and then decreased from 15 pN (Figures 2B and 3A). On the other hand, the fast mean dwell times (τ_{bf} in Equation 2) hardly depended on the load. The dwell times of slow (τ_d) and fast (τ_{df}) detachments were almost the same as τ_b and τ_{bf} , respectively, of the backward steps (Figures 2B and 3A). It is suggested that kinesin takes slow-events (forward step, backward step and detachment) and fast-events (backward step and detachment), fast-events are processes independent of the slow ones, and dwell times of slow-events depend on load but those of fast-events did not.

2.2 | Occurrence fraction of slow forward step, backward step and detachment (slow-events) and fast backward step and detachment (fast-events)

We analyzed the occurrence fractions of slow-events and fast-events to understand which reactions are predominant at each load. The fractions were calculated by the numbers (N_i and N_{if} in Equations 1 and 2) divided by the total numbers of slow-events and fast-events. At a low assisting load (–10 to 0 pN), the fraction of the forward step was almost 1, indicating that almost all steps observed were forward

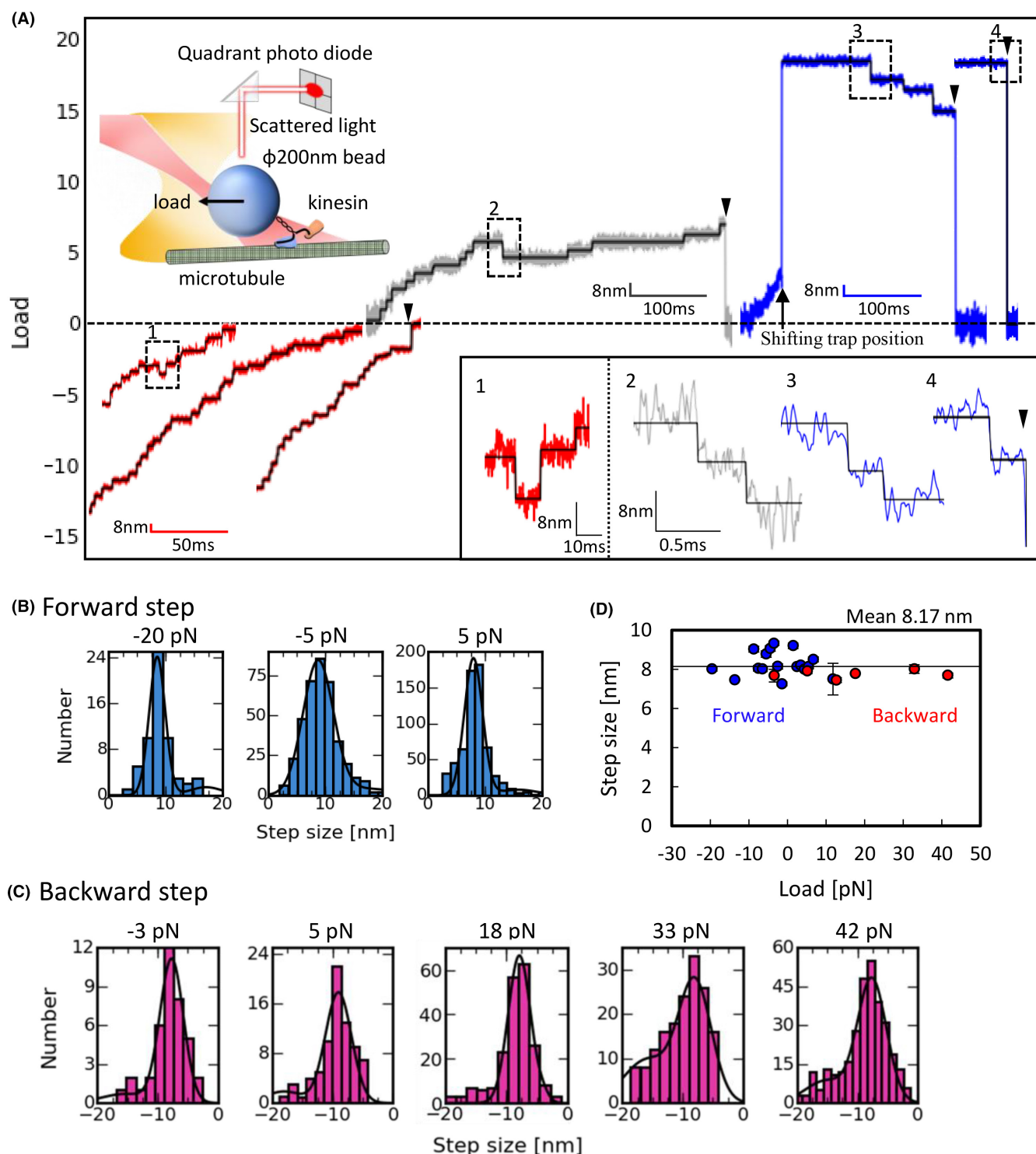


FIGURE 1 Movement of a single molecule of kinesin under loading. (A) Traces of the kinesin stepping motion under an assisting load (red) and a zero-to-stall load (gray) and under a superstall load (>13 pN). Solid lines are steps calculated by the step-finding algorithm. The backward step under an assisting load is observed at an expanded time scale (inset 1). Fast backward steps (insets 2 and 3) and detachments (inset 4) were detected at high temporal resolution. The arrow head shows detachment (>20 nm). (B, C) Histograms of the step size of the forward (B) and backward (C) steps. (D) Step size of kinesin under a wide range of loads. The straight line represents the mean of all data.

steps. The fraction of the forward step decreased with increasing assisting load, while the fraction of detachments increased (Figure 3B). The backward step was a rare event under the

assisting load, and its fraction decreased with increasing assisting load. Fast-events were not observed under the assisting load (Figure 3B).

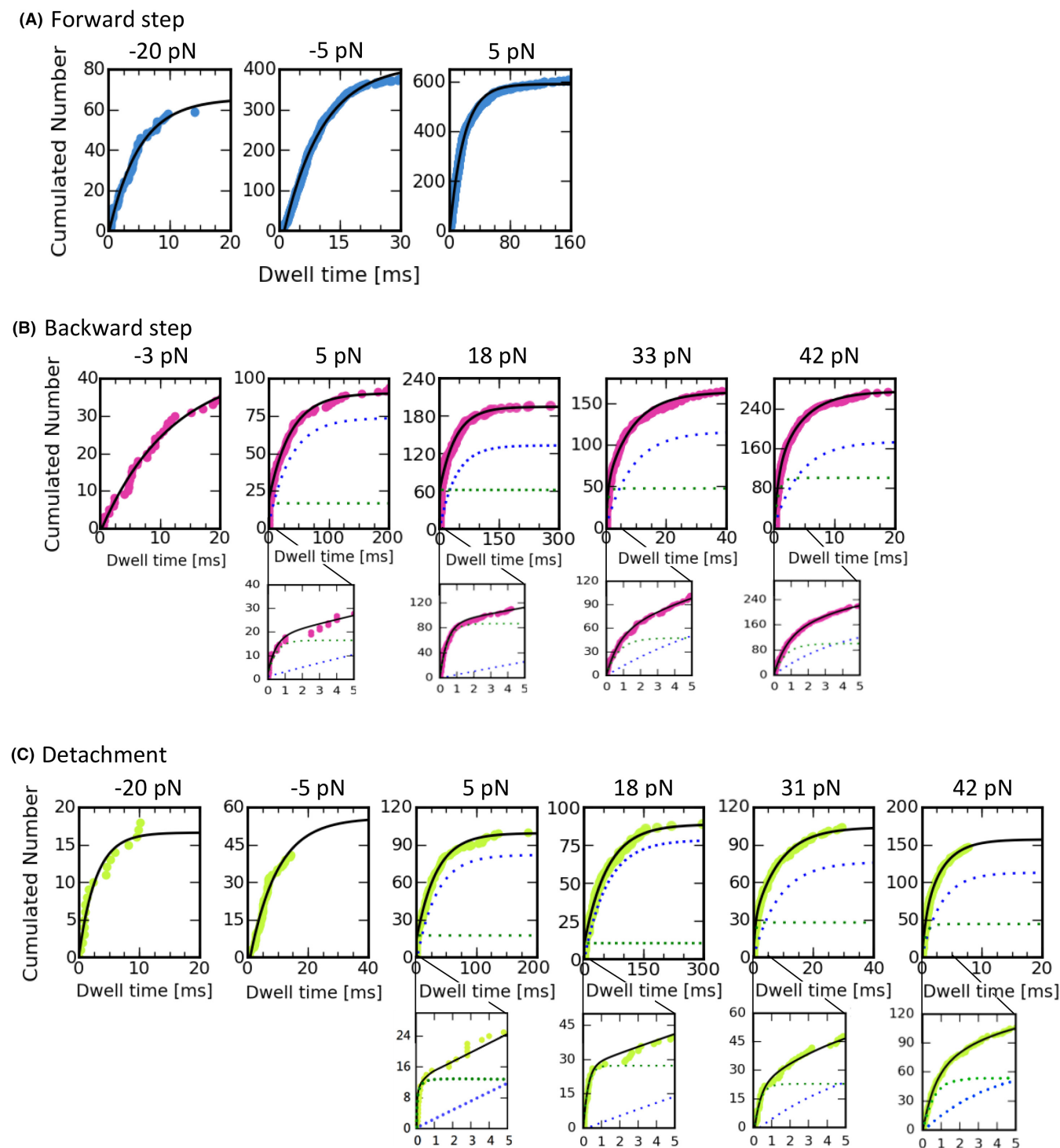
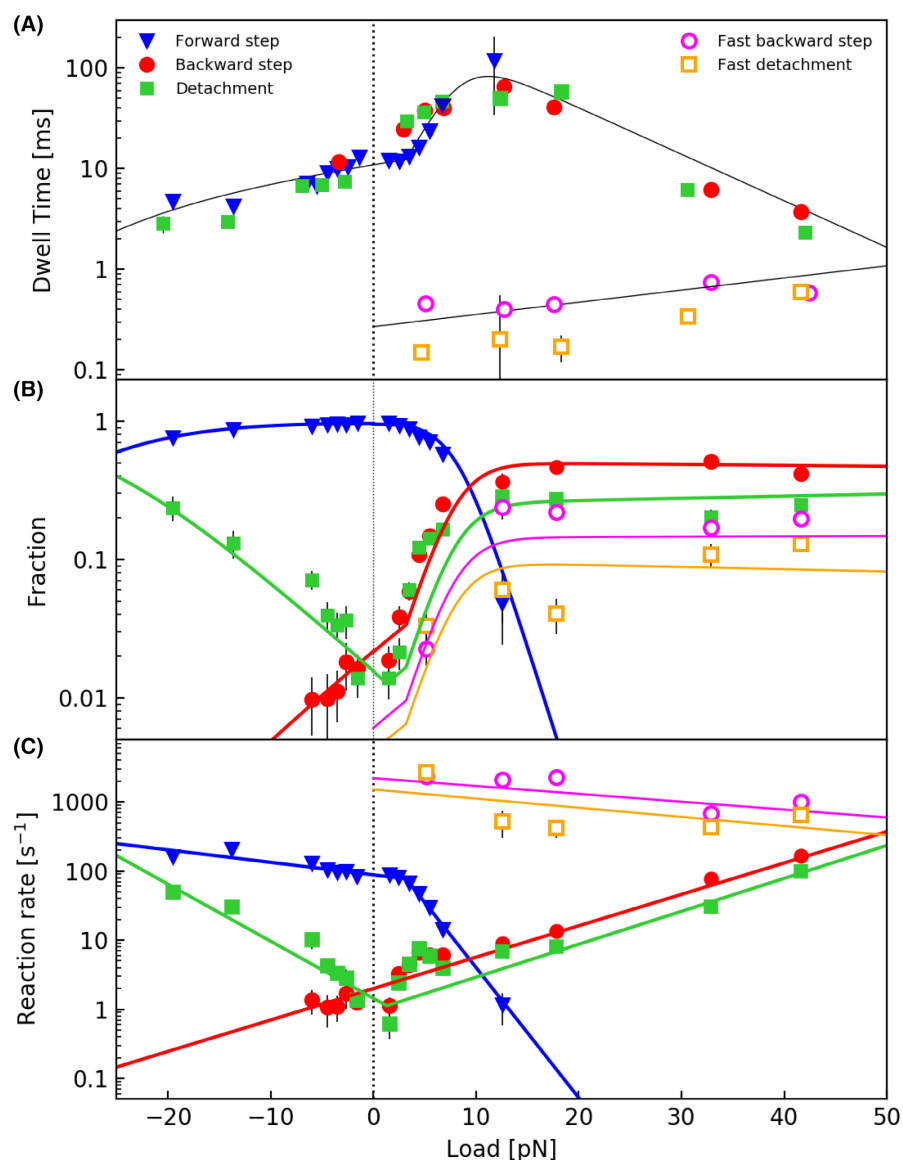


FIGURE 2 Cumulative distribution of the dwell times of the forward steps (A), backward steps (B) and detachments (C). Forward steps were fitted to Equation (1), and backward steps and detachments were fitted to Equation (2). Blue and green dotted lines indicate the first (slow steps) and second (fast steps) terms, respectively, of Equation (2). It is noted that the fitted lines do not start from $x = 0$ and $y = 0$ because of the negative term ($-n_i$) in Equations (1) and (2).

The fraction of forward steps decreased with increasing positive load, while the slow backward step and slow detachment increased (Figure 3B). The fraction of forward steps was only 4% at 10–14 pN. The fractions of slow backward steps (~ 0.5) and

slow detachments (~ 0.3) became almost constant at loads > 15 pN. Fast-events were observed at a load of 5 pN, and the occurrence fractions of the fast backward steps and fast detachments were $\sim 3\%$ of the total number of steps (Figure 3B). The

FIGURE 3 Dwell time, fraction and rate of kinesin motion. The mean dwell times (A), fractions relative to the total number of events (B) and reaction rates (C) of slow-events (forward and backward steps and detachment) and fast-events (backward step and detachment). Load-dependent reaction rates were fitted to simple exponential curves (C). The data of reaction rates in (C) were calculated from the data in (A) and (B) by using Equation (3) or Equation (4). The data points in (C) were fitted by regression lines. Theoretical curves of the dwell times (A) and fractions (B) (solid lines) were calculated from the regression lines in (C) using Equations (S16) (dwell times) and (S17 and S18) (fractions). Closed triangles, forward steps; closed circles, backward steps; closed squares, detachment; open circles, fast backward step; and open squares, fast detachment. The mean forces of slow-events and fast-events in (B) and (C) were calculated so that all forces were the same in similar force ranges. It is noted that the dwell time (~ 50 ms) near stall force is a few times shorter than that of full length of bovine and *Drosophila* kinesin-1.^{8,10}



fractions of fast-events to total steps were almost constant at loads >15 pN and decreased at 5 pN.

2.3 | Kinetic model and rate constants of slow-events

We proposed a reaction model of slow-events without fast reactions, which are reaction pathways independent of slow reactions (left side in Figure 4A). One head of the motor is bound to and the other is detached from a microtubule at state-0, and two heads are bound to the microtubule at state-1 in the forward step.^{8,10,12} The backward step is known to be not the reverse of the route $0 \rightarrow 1 \rightarrow 0$ in forward step (Figure 4A,B).^{8,10,12,14} The two-heads bound state in backward step is defined as state-2 that is different from the state-1. This irreversible model was supported by the following results. In the reversible model, ATP binds to kinesin in the forward reaction but not in the reverse

reaction; thus, the ATP concentration does not affect the reaction rate of the reverse reaction. However, the dwell time of backward steps at low ATP concentrations was almost equal to that of forward steps, indicating that the reaction is irreversible.^{8,10,12,14} Since contaminated ADP and Pi in motility solution are limited to very low concentrations (<4 mM in the presence of 1 mM ATP reagent), a reversal reaction ($0 \leftarrow 1 \leftarrow 0$) that need to binding ADP and Pi (Figure 4B) should be negligibly slow.

The dwell time of forward step is almost the same as that of backward step and detachment in all load ranges (Figure 3A). The ratios of dwell times of backward step and detachment to that of forward steps at force -3 to 12 pN were 1.22 ± 0.23 (mean $\pm$ SE) for backward step and 1.21 ± 0.32 (mean $\pm$ SE) for detachment. This indicates no significant difference between the dwell times of the forward step and other events. This suggests that all reactions of slow detachment branch approximately from state-0 (Figure 4 and Figure S2). Therefore, all forward, backward and detachment reactions are branched approximately from state-0.

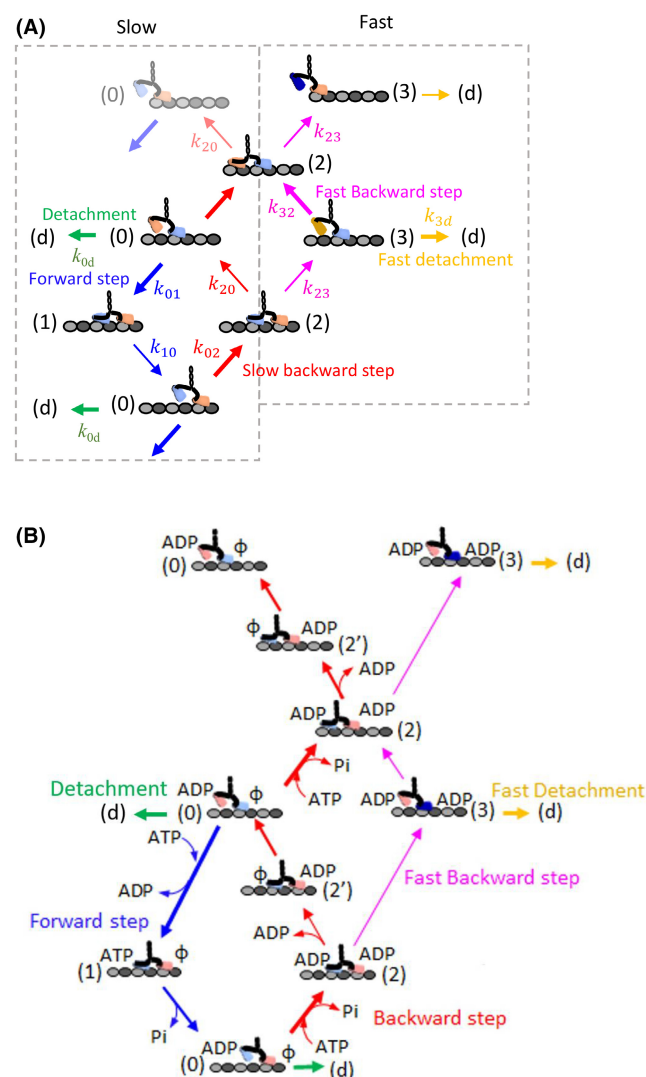


FIGURE 4 Reaction model of the stepwise movement of kinesin. (A) Kinesin moves in a hand-over-hand manner. The reaction rates with thick arrows are rate limiting. (B) Possible chemical states in the reaction model.

Which is the rate-limiting reaction, k_{01} (rate from state-0 to state-1) or k_{10} (Figure 4)? The reaction of the single head binding to a microtubule is the rate-limiting reaction at high load.¹⁵ The dwell times of slow-events fit well to the single exponential curves (Figure 2), indicating that the reactions of slow-events can be approximated as reactions with a single limiting rate. This is consistent with the previous works.^{8,16} Therefore, the reaction rates of k_{01} and k_{02} are rate limiting in the forward and backward steps, respectively, and the reaction was simplified as shown in Figure S2 using the rate constants (k_f , k_b and k_d) of slow-events.

2.4 | Reaction model including fast-events

Fast-events were observed at a load of 5–42 pN (Figure 3A). To construct a reaction model including fast-events, we analyzed the dwell

times of the initial backward steps just after loading by moving the laser trap position and of the subsequent backward steps. The cumulative dwell times of the initial backward steps were fitted to Equation (1) with a slow time constant (Figure 5A), indicating one rate limiting present in the reaction of slow-events. If k_{20} is the rate limiting (slow), k_{02} is very fast (Figure 4A). In this case, we should observe the first fast-event ($0 \rightarrow 2 \rightarrow 3$) with the dwell time of $1/k_{02} + 1/k_{23}$. This assumption is denied by the result of only slow-events. Then, if k_{02} is the rate limit, k_{20} is very fast. In this case, the route ($0 \rightarrow 2 \rightarrow 3$) becomes slow backward step, consistent with the result. Dwell times of the subsequent (second and the following) backward steps were fitted to Equation (2) with slow and fast time constants (Figure 5B). The slow time constant of the initial backward direction is approximately equal to that of the subsequent events (second backward steps and detachment). From these results, we proposed the model for fast backward stepping, in which the initial slow steps have two routes ($0 \rightarrow 2 \rightarrow 0$ and $0 \rightarrow 2 \rightarrow 3$), the transition of $0 \rightarrow 2$ is rate limiting ($k_{02} \ll k_{20}$ and $k_{02} \ll k_{23}$) for the initial steps, and the subsequent fast step is taken by two fast routes ($3 \rightarrow 2 \rightarrow 3$ and $3 \rightarrow 2 \rightarrow 0$) or two slow routes ($0 \rightarrow 2 \rightarrow 0$ and $0 \rightarrow 2 \rightarrow 3$) again (Figure 4A and Figure S2).

We also analyzed the dwell times of an initial detachment before taking the initial backward step and the subsequent detachment after it. The cumulative dwell times of the initial detachment were fitted to Equation (1), and the subsequent detachments were fitted to Equation (2) (Figure 5C,D). The slow time constant in the initial and subsequent detachments was approximately equal to that of the slow backward step (Figure 5D), suggesting that kinesin detaches from state-0 in the initial detachment (Figure 4A). The fast time constant (k_{df}) of the subsequent detachment is approximately equal to that of the backward step, suggesting that reactions of the detachment and backward steps branch at state-3.

Which of reaction $3 \rightarrow 2$ and sequential reaction $2 \rightarrow 3$ is rate limiting in the fast backward reaction of ($3 \rightarrow 2 \rightarrow 3$)? The dwell time and fraction of the fast backward step ($3 \rightarrow 2 \rightarrow 3$) are not very different from those of fast detachment ($3 \rightarrow d$). In addition, the rate of the fast backward step is approximately a single rate constant because the cumulative dwell time fits two independent single exponentials (Equation 2), suggesting that the reaction of $3 \rightarrow 2$ is rate limiting. The reactions of fast-events are approximately represented in Figure S2 using the single rate constants k_{bf} and k_{df} of the fast backward step and detachment. The reversal reaction of fast backward step ($3 \leftarrow 2$ and then $2 \leftarrow 3$) observed at high load is assumed to be slow. The reason is that the reversal reaction of fast backward step that generates the forward step needs the energy at positive load, but there is no energy source at the reversal reaction from ADP-binding to ADP binding state (Figure 4B).¹⁶ Therefore, the reversal reaction rate of fast backward step will be negligible compared with that of the forward reaction.

To understand whether ATP binds in the fast reaction, the dwell times of slow-events and fast-events were analyzed. The cumulative numbers of dwell times at ~ 18 pN fit well to Equation (2) (Figure 5E,F). The dwell times of slow-events at an ATP concentration of 10 μ M were approximately 200 ms slower than that at a high ATP

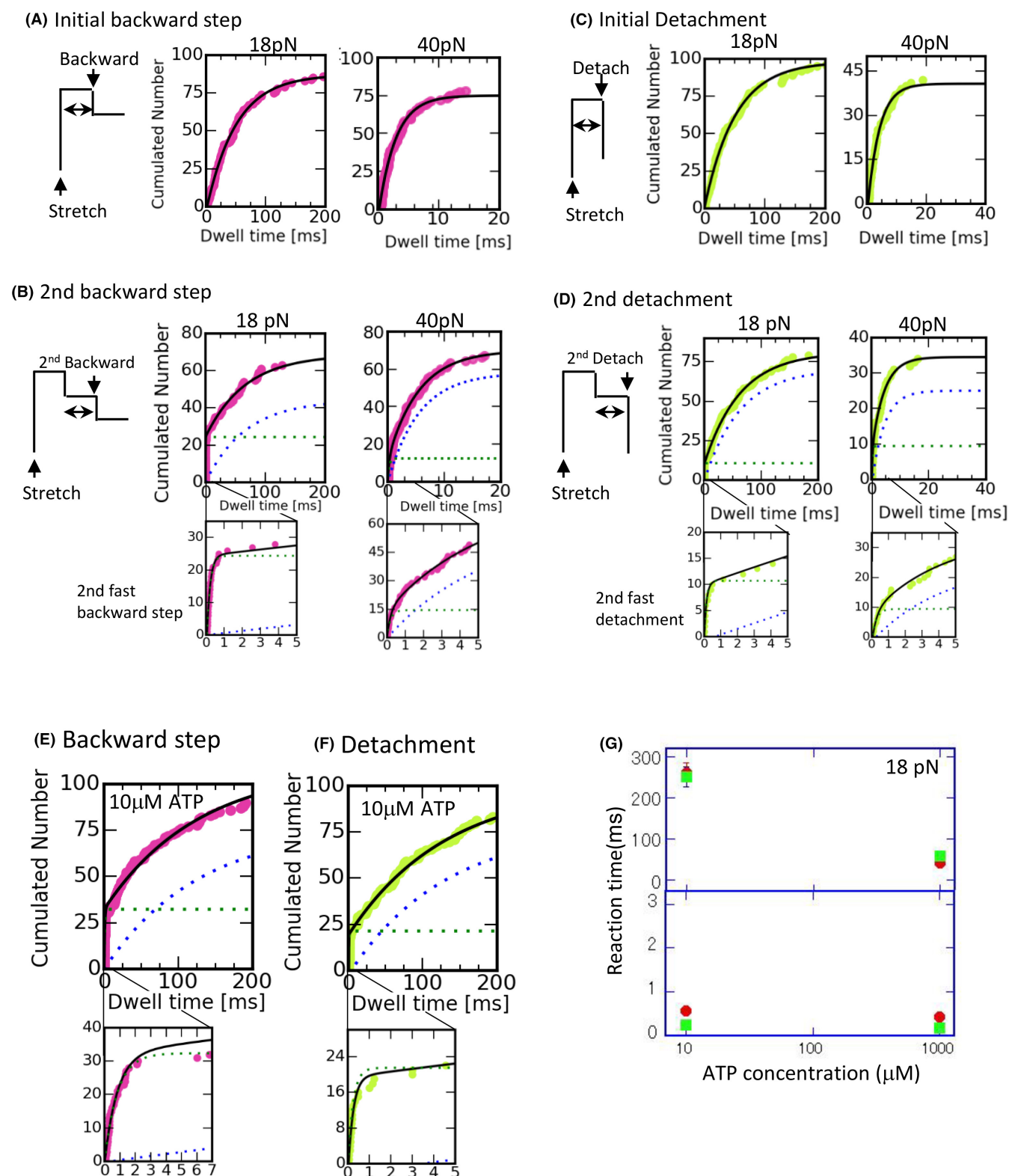


FIGURE 5 Fast backward step and detachment. Dwell time distribution of the initial (first) backward step and detachment (A), second backward step (B), initial detachment before the initial backward step (C) and second detachment after that (D). In the schematic drawings, the double arrows indicate the dwell times measured in each figure after the kinesin was stretched by moving the position of the laser trap. (E, F) Cumulative numbers of backward steps and detachments at low ATP concentrations (10 μM ATP), fitted by Equation (2). (G) Dwell times of slow-events and fast-events at high and low ATP concentrations. The symbols are the same as those in Figure 3.

TABLE 1 Parameters of the rate constants.

	λ_j [s^{-1}]	L_j [pN]	d_j [nm]
k_{f+}	77.3 ± 0.9	3.19 ± 0.05	-1.78 ± 0.04
k_{f-}			-0.17 ± 0.01
k_b	1.99 ± 0.01	0	0.44 ± 0.01
k_{d+}	1.12 ± 0.07	1.32 ± 0.55	0.45 ± 0.01
k_{d-}			-0.78 ± 0.05
k_{fb}	2200 ± 800	0	0.11 ± 0.04
k_{fd}	1510 ± 690	0	0.13 ± 0.06

concentration. However, the dwell times of fast-events at 10 μ M ATP were close to that at 1 mM ATP (Figure 5G). These results indicate that ATP molecules do not bind to kinesin during fast-events.

2.5 | Rate constants of slow-events and fast-events

How can the rate constants of slow-events and fast-events be calculated analytically from experimental results? The rates of the branched reactions are calculated from the fraction (f) and dwell time (τ) using the following equations (Equations S20 and S21):

$$k_i = \frac{1}{\tau_s} \frac{f_i}{f_f + f_b + f_d} \quad (i = f, b, d) \quad (3)$$

$$k_j = \frac{1}{\tau_{fa}} \frac{f_j}{f_{bf} + f_{df}} \quad (j = bf, df) \quad (4)$$

where $1/\tau_s$ and $1/\tau_{fa}$ are calculated to be $(\tau_f^{-1} + \tau_b^{-1} + \tau_d^{-1})$ and $(\tau_{bf}^{-1} + \tau_{df}^{-1})$, respectively. Five rate constants calculated from the fraction and dwell time are plotted in Figure 3C.

The logarithm of the rate constants was fitted with a piecewise-linear curve (Figure 3C). The slopes of $\log k_f$ were gentle under an assisting load and steeper under a hindering load. The logarithm of k_d increased linearly with increasing the assisting and hindering loads, similar to a V-shape. The logarithm of k_b increased monotonically with the load. The slopes of $\log k_{bf}$ and $\log k_{df}$ were very gentle. From these linear relations, the reaction rates plotted in Figure 3C were fitted to a single exponential function,

$$k_j(L) = \lambda_j \exp \left[\frac{d_j(L - L_j)}{k_B T} \right] \quad (j = f_+, f_-, b, d_+, d_-, bf \text{ and } df) \quad (5)$$

where d_j is a characteristic distance, λ_j is the rate constant, L is the applied load and L_j is the load at the intersection of two rates in the slow forward step and detachment. The suffix “−” is for $L < L_j$, and the suffix “+” is for $L \geq L_j$. $k_B T$ is the thermal energy. The constants in Equation (5) are summarized in Table 1. L_d (1.32 ± 0.55 pN, mean $\pm$ SE) for detachment was close to zero, but L_f (3.19 ± 0.05 pN) was not.

After we obtained the parameters (Table 1) in Equation (5) for slow-events and fast-events, the dwell time and the fraction (Figure 3A,B) were then calculated from Equations (S16–S18), which were calculated from differential Equations (S1–S5). Both the dwell time and calculated fraction were consistent with the experimental data, supporting the validity of the differential equations (Figure 3A,B).

3 | DISCUSSION

In this study, we investigated the stepping kinetics of single kinesin molecules under very wide range of loads with high temporal resolution for the first time. We found new reactions of fast-events that have a mean dwell time of ~ 0.4 ms. A simplified reaction model with four states and five rate constants was proposed. All five rate constants were fitted to a single exponential under assisting and hindering loads. The reactions containing the five rate constants were analytically solved (Figure 3C and Equations S20 and S21). We proved that the simplified model (Figure 4 and Figure S2) is sufficient to explain the movement of kinesin under a load.

3.1 | Eight-nanometer slow and fast steps

We successfully detected the stepping motion under a wide range of loads from -20 to 42 pN. Kinesin showed 8-nm steps even at a 42 pN load, which is five times higher than the stall force, indicating that hand-over-hand motion with a step size of 8 nm is a robust feature of kinesin's mechanochemical reaction. Regular 4-nm steps observed by Sudhakar et al.¹⁶ were not detected in the present work. In the case of recombinant yeast dynein, the load-induced step size is variable (4–24 nm) and is caused by inchworm motion or compact and extended conformations of the dimer.¹⁷ The robustness of kinesin's step size is regulated by the length of the neck linker.¹⁸ These suggest that the neck linker is short and has high stiffness,¹⁹ and thus the tethered head cannot reach a binding site at distances larger than 8 nm from the bound head even under a superstall load.

A fast backward step and detachment with a dwell time < 1 ms were detected by the improved temporal resolution of displacement measurements. A slow backward step, followed by a fast backward step, was apparently detected as a 16-nm step at low temporal resolution. In fact, slow 8- and 16-nm backward steps were reported previously at 3–8 pN with a lower temporal resolution of ~ 2 ms or longer^{8,14} and were observed in the experimental trace at 10–12 pN.¹⁰ The number fraction of fast to slow backward steps observed in the present work ($\sim 15\%$ at 3–8 pN, Figure 3B) is almost the same as that of -16 -nm steps reported previously ($\sim 13\%$ at 2–8 pN⁸ and $\sim 10\%$ ¹⁴). Therefore, we concluded that most of the previously observed 16-nm steps consisted of slow and consecutive fast 8-nm steps. The Cross group regarded longer backward steps as slips¹⁴ because they were too fast to consist of multiple tightly coupled 8-nm steps. This agrees with our model that fast 8-nm

backward steps do not involve a biochemical state transition (Figure 4B). The backward slip-steps detected by Sudhakar et al.¹⁶ might be the same events as fast backward steps detected here. In our work, 10%–20% of 16-nm backward steps was not resolved into double 8-nm steps. The dwell time of fast backward step was ~ 0.4 ms. Assuming the cumulated number of dwell time increases exponentially with dwell time, 18% of the total number of dwell times is calculated to be shorter than 0.1 ms. 18% is within the experimental data of 10%–20%. Therefore, 10%–20% of 16-nm backward steps, which were not resolved into double 8-nm steps, will be undetectable fast double 8-nm steps occurred within ~ 0.1 ms.¹⁶

3.2 | Chemical states in the reaction model

Which chemical state of kinesin heads is the rate-limiting one (Figure 4)? State-0 of a single-head binding is a rate-limiting state in the present model. This is supported by a previous report that a kinesin head linked with a DNA bead spends approximately half of its stepping cycle in the unbound state (“overshoot dwell” in the literature),¹⁵ suggesting that either head spends most of the stepping cycle in the unbound state under high load. The rate limiting of the stepping reaction is thought to be an ATP waiting state in which the neck linker on a bound apo-head is supposed to be undocked (Figure 4B).¹⁵ The rate constant of neck linker docking is load-dependent,²⁰ suggesting that the load dependency of rate-limiting k_{01} is predominantly correlated to that of neck linker docking.

Isojima's group suggested that two-head bound state is rate limiting at no load by measuring the fluctuation of kinesin head labeled with nanogold.²¹ As an option, we constructed the reaction model in which the rate limiting is two-head bound state (Figure S3) and solved the model to analytically obtain the reaction rates (Equations S39–S43 and S46). At low and assisting load, including no load like Isojima's work, the fraction of fast-events and backward steps are negligibly small, so the reaction model became very simple (Figure S3C) and was solved easily to obtain the rates. The optional model may be needed at narrow load region inside of wide range we studied.

The fast backward steps are almost independent of the ATP concentration at ~ 18 pN (Figure 5G). The reaction times (k_{bf}^{-1}) of the fast backward step at 10 μ M ATP were too short (~ 0.5 ms) to bind ATP in the process of the backward step (Figure 5G). In addition, because ADP and Pi were not added to the motility solution (contaminated ADP and Pi < 40 nM in 10 μ M ATP), ADP and Pi rarely bonded to the head in the fast backward step. To satisfy these conditions, we propose a model of a fast backward step in which both heads bind to microtubules in the ADP state (state-2), the leading head dissociates quickly under a load (state-3), and the dissociated head binds to the microtubule (state-2) and returns to state-3 without binding ATP, ADP and Pi (Figure 4B). Other possibilities that the state-2 is ADPPi state will be ruled out because kinesin head at ADPPi did not dissociate very fast (< 0.1 ms) from microtubule.^{14,16} Therefore, the reactions (states $3 \rightarrow 2 \rightarrow 3$ and $3 \rightarrow 2 \rightarrow 0$) in the fast backward step were the

uncoupling step without utilizing ATP hydrolysis (Figure 4B and Figure S2). The rate-limiting process in the uncoupling cycle will be the binding of the ADP-bound head to the microtubule (state-3 to state-2) because the ADP state is weakly binding.²²

In our model, we assumed that slow-events are the reactions branching from the common state-0. Actually, the dwell times of backward steps and detachment at -3 to 12 pN were not significantly different from those of forward steps; the ratios (backward/forward and detachment/forward) of dwell time were 1.22 ± 0.23 and 1.21 ± 0.32 . These results are consistent with the ratio of ~ 1.2 obtained at GDP-Taxol-microtubules by Toleikis' work.¹⁴ In their work, the reaction route of forwards steps branch at common state (their state-2) from that of backward steps and detachments, and then backward steps branches from detachments at the next-state (state-3) following to state-2.¹⁴ When the reaction from the state-2 to the state-3 is very fast, the forward step is approximated to branch from the same state of backward steps and detachment.

3.3 | Two characteristic distances in the rates of forward steps and detachments

The characteristic distances in the rates of forward steps and detachments were changed at low load. The distances of slow detachment were negative (d_{d-}) under an assisting load and positive (d_{d+}) under a hindering load, supporting previous results obtained in the presence of AMPPMP.^{22,23} The absolute value of d_{d-} was significantly different from d_{d+} , suggesting that the asymmetry of kinesin binding sites to microtubules contributes to the difference in the characteristic distances. The angstrom order of the characteristic distance of d_{f-} (-0.17 nm) is close to the previous result of 0 distance.¹⁰

To understand the characteristic distance, the energy diagram that is defined by displacement on x-axis and free energy on y-axis with an activated state between stable states was used often.²⁴ Assuming the free energy diagram of the reaction is tilted by the external force (load), the load did not affect so much on rate of the reaction from stable to activated states when the distance between stable and activated states is very short (or potential slope is steep).²⁴ The concept of distance in the energy diagram is similar to the characteristic distance. It is suggested that the distance in the energy diagram is short at assisting load and gentle at hindering load ($d_{f+} = -1.78$ nm).^{23,24}

3.4 | Simple reaction models of kinesin

The dwell time and fraction of slow-events calculated from differential equations were expressed as a complicated equation (Equations S16–S18). In addition, the logarithm of k_f and k_d changed the slopes at 3.2 and 1.3 pN, so Equations (S16–S18) need to be calculated at three regions: < 1.3 , 1.3 – 3.2 and > 3.2 pN, respectively. Therefore, a simple model of kinesin stepping without

dividing the calculation regions is useful for understanding the mechanochemical cycle of molecular motors. A unified model, in which reverse and detachment reactions were neglected, for the processive motor proteins kinesin-1, dynein-1 and myosin-V is simple and predicts the force dependence of the dwell time and step ratio of backward to forward steps under a wide range of loads.¹² In the model, the rate of forward steps at the assisting load was assumed to be constant because the dwell time of the forward steps at the assisting load in the previous model did not depend on the load based on the experimental results.^{10,25} It is likely that the dwell time of the assisting load in the previous experiment¹⁰ was overestimated because most short dwell times <4 ms were not included when calculating the mean dwell time. Here, the temporal resolution was improved to ~0.1 ms, and the mean dwell time was calculated accurately by fitting the cumulative dwell time (Figure 2). Therefore, the present result that the dwell time decreased with increasing assisting load is accurate. In addition, detachment was not included in the unified model because the dwell time of detachment was measured only within a narrow range of loads (2–8 pN).^{8,9} In this work, the detachment was measured over the whole range of loads (–20 to 42 pN) (Figure 3A). In the present result, we are able to express the dwell time and step ratio by simple theoretical equations, including the rates of forward and backward steps and the detachment over a wide range of loads, as in previous work.¹²

To calculate the dwell time and step ratio by simple equations such as Sasaki's model,¹² the forward step rate (k_{01}) and the detachment rate (k_{0d}) were approximately expressed as the sum of two exponential terms (Equations 6 and 8). To simplify the model, fast-events were neglected because its fraction was less than 30%. The characteristic distance of the detachment at the hindering load was the same as that of the backward step (d_2) (Figure 3C). Therefore, the rates are as follows:

$$\frac{1}{k_{01}} = \frac{1}{\kappa_0} \exp\left(\frac{d_0 L}{k_B T}\right) + \frac{1}{\kappa_1} \exp\left(\frac{d_1 L}{k_B T}\right) \quad (6)$$

$$k_{02} = \kappa_2 \exp\left(\frac{d_2 L}{k_B T}\right) \quad (7)$$

$$k_{0d} = \kappa_d \left[\exp\left(\frac{d_d L}{k_B T}\right) + \exp\left(\frac{d_2 L}{k_B T}\right) \right] \quad (8)$$

Then, the dwell time and step ratio were calculated via Equations (S50–S52) (Figure S4). The load dependence of the step ratio was gentle at <4 pN and then steep at >6 pN (Figure S4A). The dwell times increased steeply at a high assisting load, gently at a low assisting load and steeply at 4 < load < 8 pN; additionally, they decreased gently at a superstall load (Figure S4B). The obtained characteristic distances, namely, d_0 (–0.18 nm), d_2 (0.42 nm) and d_d (–0.86 nm), are reasonably close to d_{f-} (–0.17 nm), d_b (0.44 nm) and d_{d-} (–0.78 nm), respectively (Figure S3B). On the other hand, d_1 (–3.56 nm) was approximately twice that of d_{f+} (–1.78 nm) because Equation (6) is different from Equation (5).

3.5 | Application of rate constants

Rate constants determined under a wide range of loads are useful for understanding the motility of single and multiple kinesin molecules. The velocity of single kinesin molecules was calculated from the step size and rate constants (Figure S5A, Equation S53). The velocity is 707 nm/s at no load, zero at a stall load of 9.1 pN and the velocity increased with the increase in assisting load, consistent with previous reports.^{6–8,10,26} The mean run length of kinesin was also calculated by taking the velocity divided by the detachment rate (Figure S5B, Equation S54). The maximum run length of 595 nm is consistent with previous studies.^{20,27}

The reaction rates are important for understanding the movement of multiple motor systems. In the case of tug-of-war, in which many molecules of dynein and a small number of kinesin molecules attach to a vesicle, kinesin molecules will take a large load from many dynein molecules.^{28–30} Kinesin is able to move faster in the backward direction in the presence of fast-events, and then dynein moves faster at a lower load. In other cases in which a small number of slow-type kinesin molecules interact with many fast kinesins during mitosis,³¹ slow kinesin moves under an assisting load by fast kinesin movement. The present work is important not only for understanding the motile mechanism of single kinesin molecules but also for simulating the movement of multimolecular systems.

4 | MATERIALS AND METHODS

4.1 | Preparation of proteins

Double-headed kinesin-1 (KIF5a, mouse), 490 amino acids, was fused to biotin-tag and HIS-tag at its C-terminus. KIF5a was expressed in BL21 competent *Escherichia coli* and purified by Ni-NTA affinity chromatography and microtubule affinity purification.³² Tubulin was obtained from bovine brains and labeled with tetramethylrhodamine succinimidyl ester. The microtubule was prepared by polymerizing tubulin purified from porcine brain at 37°C for 30 min.³³

4.2 | Optical tweezers

We used the optical tweezers system described previously with some modifications to reduce noise.^{3,33} The optical table was equipped with an active vibration isolation system (AVI-350M/LP, Herz Co., Ltd.). The noise from optical parts on the table and electrical motors outside of the table was minimized. The intensity of the bead image on the quadrant photodiode was improved to approximately three times that in the previous system by focusing on diagonal illumination. As a result, the standard deviation of the displacement noise of 200 nm diameter beads, fixed on the coverslip, was reduced to 0.3 from 0.6³ and 1–2 nm³⁴ in the previous systems at a frequency range of 2 Hz to 30 kHz.

A bead was trapped in the center of a focused infrared laser beam (1064 nm, 800 mW, Showa Optonics Co., Ltd.) by an objective lens (NA 1.40, Plan Apo N ×60 OSC, Olympus). The beads were diagonally

illuminated by a focused red laser beam (690 nm, 50 mW, CrystaLaser), and the scattered light was collected through an objective lens (setting the aperture diaphragm to an NA of 0.5, Plan Apo $\times 100$, Olympus). Dark-field images were projected onto a quadrant photodiode detector.

4.3 | Single-molecule measurement by optical tweezers

The microtubule was immobilized on a glass surface by modifying the method described previously.³³ Coverslips with sizes of $24 \times 32 \text{ mm}^2$ and $18 \times 18 \text{ mm}^2$ were washed with a plasma cleaner (PDC-32G, Harrick Plasma Co., Ltd.). Coverslips with a size of $24 \times 32 \text{ mm}^2$ were further treated with APTES (aminopropyltriethoxysilane, ShinEtsu Co., Ltd.) for 40 min. APTES on the glass surface was washed off with ultrapure water. Flow chambers with a volume of 2–3 μL were made by using the APTES-coated large coverslip, double-sided tape (30 μm in thickness) and plasma-cleaned small coverslip. A 20% glutaraldehyde solution (Wako) was introduced into the chamber and incubated for 30 min. Each chamber was washed with 100 μL of water. The microtubule was flowed into the chamber and fixed onto the coverslip by incubation for 1 min. Unbinding microtubules were washed out with 10 μL of buffer (80 mM PIPES, 2 mM MgSO_4 , 1 mM EGTA, 20 μM Taxol, pH 7.2). The microtubule-binding chambers were kept in a humidified box for several hours.

Fluorescent beads of 200 and 400 nm were coated with streptavidin and stored at -80°C .³ The streptavidin beads were mixed with biotinylated kinesin at a molar ratio of 1:20 to 1:40 (bead: kinesin), where 10%–30% of the beads showed motility. Kinesin beads were diluted with a motility solution (25 mM PIPES, 25 mM potassium acetate, 2 mM MgSO_4 , 1 mM EGTA, 1 mg mL^{-1} casein, 20 mM glucose, 10 $\mu\text{g mL}^{-1}$ catalase, 20 $\mu\text{g mL}^{-1}$ glucose oxidase, 1% 2-mercaptoethanol, 10 μM paclitaxel and 10 μM to 1 mM ATP at pH 7.2) at $25 \pm 1^\circ\text{C}$. It is noted that we used 25 mM PIPES to obtain slightly higher affinity to microtubule than 80 mM PIPES (BRB-80).^{6,8}

The trap stiffnesses were 0.090 for 200-nm beads and 0.25 pN nm^{-1} for 400-nm beads. The 400-nm beads were used for experiments at large loads of ≥ 25 or ≤ -17 pN. The position of the optical trap and bead was shifted by an angle change in a custom-made piezo mirror at kinesin forces reaching ~ 2 pN to generate the assisting load and the superstall load. The position change was terminated within 100 μs after the movement. The microtubule parallel to the direction of position change of bead was selected to keep the bead on microtubule.

The displacement was recorded at a sampling rate of 100 kHz (10 μs /sampling) for all load ranges. The roll-off frequency of the amplifier for detecting the displacement was 30 kHz (or a time constant of 5.3 μs), which is sufficient for 10 μs /sampling. Some data at a load of 0–8 pN were recorded at a sampling rate of 20 kHz. The recorded data were further treated with a low-pass filter with a corner frequency of 1 kHz (time constant of 0.16 ms) under loads from 0 to 8 pN and 2 kHz (0.08 ms) and under loads < 0 and > 10 pN to detect slow steps. The recorded data at a sampling rate of 100 kHz were analyzed without digital filtering to detect the fast step.

Step sizes and dwell times were analyzed by a step-finding algorithm that was programmed by Python.³⁵ All the dwell times were longer than 50–100 μs . The step sizes were calculated by multiplying the observed step sizes by the attenuation factor (~ 1.25 at absolute load < 3 pN, ~ 1.1 at > 7 pN) calculated from the trap stiffness and compliances of the linkage.⁶

The performance of the step-finding algorithm was tested as follows. We made the displacement of the artificial steps with sampling rate of 100 kHz (10 μs /point), the size of 8.2 nm and the mean dwell time of ~ 0.4 ms, which were taken from the present results of sampling rate, step size (Figure 1D) and dwell time of fast backward steps (Figure 3A). Noise of 2.38 and 2.81 nm (standard deviation), which were close to mean noises at force of 4–5 and 6–7 pN, respectively, was added to the displacement and then the displacement with noise was processed by the low-pass-filter with bandwidth of 30 kHz; in the present work, corner frequency of displacement measured at force > 4 pN was higher than the frequency (~ 30 kHz) of the amplifier. The filtered displacements were analyzed by the step-finding algorithm. The performance (P) was calculated from the numbers of correctly found steps (N_c), undetected steps (N_u) and misdetected steps (N_m) using the equation of $P = N_c/(N_c + N_u + N_m)$.

The results of step finding and the performance are shown in Figure S6. Step finding algorithm found the steps very accurately at the noise of 2.38 nm after processing low-pass filter at 30 kHz (Figure S6A). The performances at dwell times of 10 and 20 μs were zero, so the performances in histograms were started from 30 μs (Figure S6B,C). The performances were increased with increase in the dwell time. The performances were $> 90\%$ at dwell time $\geq 100 \mu\text{s}$ (noise of 2.38 and 2.81 nm) and $\geq 77\%$ at dwell time $\geq 50 \mu\text{s}$ (2.38 nm). The dwell time ≥ 100 and $\geq 50 \mu\text{s}$ at force < 10 and > 10 pN, respectively, were analyzed in the present experiment because the noise of experimental displacement decreased with the increase in absolute values of loads.

4.4 | Analysis of the dwell time

Akaike's information criterion was calculated to determine which Equation (1) or Equation (2) is favored for the cumulative number of dwell times.³⁶

AUTHOR CONTRIBUTIONS

Yuichi Kondo and Hideo Higuchi designed the project. Yuichi Kondo conducted the experiments and the data analyses. Kazuo Sasaki, Yuichi Kondo and Hideo Higuchi constructed the theoretical equations. Hideo Higuchi wrote the paper with input from Kazuo Sasaki.

ACKNOWLEDGMENTS

The authors thank Motoshi Kaya, University of Tokyo, for helping the analysis of step finding to measure the performance. This work was supported by Grants-in-Aid for Scientific Research from Japan MEXT (HH, JP16H04773 and 19H03189).

CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

PEER REVIEW

The peer review history for this article is available at <https://www.webofscience.com/api/gateway/wos/peer-review/10.1111/tra.12909>.

DATA AVAILABILITY STATEMENT

The datasets used and/or analyzed during the current study available from the corresponding author on reasonable request.

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SUPPORTING INFORMATION

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How to cite this article: Kondo Y, Sasaki K, Higuchi H. Fast backward steps and detachment of single kinesin molecules measured under a wide range of loads. *Traffic*. 2023;24(10):463-474. doi:10.1111/tra.12909